

UNIT-5

Planning and Learning

Learning in Artificial Intelligence

Machine Learning is a branch of AI that **enables systems to improve performance** automatically through **experience** and data without being explicitly programmed.

Learning is the process through which a system acquires knowledge or skills and improves its performance over time.

“Any **change in a system** that allows it to **perform better the next time on the same task.**”

Humans learn from **experience, observation, and practice.**

Similarly, intelligent agents in **AI learn from data, examples, and interaction** with the environment.

Definition-

A program learns from:

- **Experience (E)**
- With respect to some **Task (T)**
- And improves according to **Performance Measure (P)**

Example

- A spam filter learns to identify spam emails.
- A chess program learns better moves after playing many games.
- A recommendation system learns user preferences over time.

Components of Learning

A learning system generally contains:

Component	Description
Performance Element	Performs the task
Learning Element	Improves performance
Critic	Gives feedback
Problem Generator	Suggests exploratory actions

Forms of Learning

Rote Learning

Rote learning means **memorizing facts** or **solutions** without understanding.

Characteristics

- In rote learning, the system **stores previously solved problems** or **facts** directly into memory for future use.
- The stored information can be **retrieved very quickly** whenever the same problem appears again.
- This type of learning **does not involve reasoning** or **understanding**; the system simply memorizes and recalls information.

Examples

- A **dictionary** lookup is an example of rote learning because the meaning of words is stored and retrieved directly when required.
- Memorizing multiplication tables is another example, where answers are remembered without performing calculations each time.

Advantages

- Rote learning is **simple to implement** because it mainly depends on storing and retrieving information.
- It **provides quick responses** since the solution is already available in memory and does not need further computation.

Disadvantages

- Rote learning requires a **large amount of memory** to store all possible problems and their solutions.
- It cannot **generalize knowledge** to **solve new or unseen problems** because it only works for previously stored cases.

Learning by Taking Advice

The agent **learns from instructions** provided by humans or other systems.

Advantages

- Learning by taking advice enables the **system to learn quickly** because knowledge is directly provided by an expert.
- It **reduces the search effort** of the system since the learner **does not need to discover all solutions** through trial and error.

Disadvantages

- This type of learning **depends heavily on the quality and correctness** of the advice provided.
- If the **advice given is incorrect** or incomplete, the **system may learn wrong rules** and produce inaccurate results.

Learning from Examples (Inductive Learning)

The **system learns general rules** from specific examples.

Example

Example	Output
Email contains “lottery”	Spam
Email contains “discount”	Spam
Email from office	Not Spam

The system induces rules for spam detection.

Here, Specific observations → General rule

Learning by Analogy

The system **solves a new problem using similarities** with previously solved problems.

Example

If a robot knows how to navigate one building, it may apply similar strategies in another building.

Explanation-Based Learning (EBL)

The system **learns by analyzing** why an example is correct.

Steps

1. Observe example
2. Explain why it works
3. Generalize explanation

Example

A student understands why a mathematical formula works instead of memorizing it.

Advantages

- Efficient learning
- Reduces future computation

Discovery Learning

The system **discovers new rules or concepts** independently.

Example

A scientific AI system discovering relationships in experimental data.

Features

- Exploratory learning
- Generates new knowledge

Reinforcement Learning

An agent **learns through rewards and punishments**.

Process

- Agent performs action
- Environment gives feedback
- Positive reward → encourage action
- Negative reward → discourage action

Example

- Robot navigation
- Game playing
- Self-driving cars

Theory of Learning

Learning theory studies **how machines or intelligent systems acquire knowledge and improve their performance** through experience. It explains the principles and methods that make learning possible in Artificial Intelligence systems.

- Learning theory **studies how the learning process occurs** in intelligent systems and how knowledge is acquired from data or experience.
- It **examines the conditions** required for successful learning so that the system can make accurate predictions and decisions.
- It also **analyzes the efficiency of learning** algorithms in terms of time, memory, and the amount of training data required for learning.

The **main goal** is to understand:

“How can a machine learn correctly from limited experience?”

Elements of Learning Theory

Element	Description
Training Data	Examples used for learning
Hypothesis	Candidate rule/function
Target Function	Actual correct function
Error Measure	Difference between prediction and actual output
Generalization	Ability to perform well on unseen data

Different Terms/ Concept in Theory of Learning

Generalization

A learning system should not only **memorize training data** but also **perform correctly** on new unseen examples.

Example -A student who memorizes questions may fail in new questions.

A good learner:

- Understands patterns

- Applies knowledge to new situations

Bias in Learning

Bias refers to **assumptions made by a learning algorithm** to generalize beyond training data.

A learning algorithm cannot examine every possible hypothesis, so it uses certain assumptions called bias.

Types of Bias

Type	Meaning
Restriction Bias	Limits possible hypotheses
Preference Bias	Prefers some hypotheses over others

Bias helps:

- reduce complexity,
- speed up learning,
- and improve prediction.

Without bias:

- learning becomes impossible because too many hypotheses exist.

Overfitting and Underfitting

Overfitting

Model learns noise and memorizes training data.

As a result-

Poor performance on new data.

Underfitting

Model is too **simple** and **fails to learn patterns**.

As a result:

- the model performs poorly on both training data and testing data.

PAC Learning

PAC = **P**robably **A**pproximately **C**orrect

PAC learning is a **mathematical framework** introduced by Leslie Valiant for studying machine learning.

It explains: Under what conditions can a machine learn efficiently?

Basic Idea of PAC Learning

A learner is successful if it learns:

- **Probably** → with high probability
 - **Approximately Correct** → close to correct answer
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- “**Probably**” means that the learning algorithm should **produce a correct or near-correct hypothesis** with a **high probability of success**.
 - “**Approximately Correct**” means that the learned hypothesis **does not need to be perfectly correct**, but it should be sufficiently **close to the actual target concept** with only a **small acceptable error**.
 - Instead of demanding perfect learning, PAC learning **allows small errors** because achieving absolute correctness **may not always be practical in real-world problems**.
 - PAC learning **provides a mathematical framework** to determine whether a **concept can be learned efficiently** using a **limited number of training examples**.

Important Terms in PAC Learning

Term	Meaning
Concept Class	Set of possible target concepts
Hypothesis	Predicted concept
Error	Difference from target concept
ϵ (epsilon)	maximum acceptable error
δ (delta)	maximum acceptable probability that the learner fails

PAC Definition

A hypothesis h is considered PAC-learned if:

- Error of the hypothesis is less than or equal to ϵ
- Probability that the hypothesis fails is less than or equal to δ

Mathematically:

$$P(\text{error}(h) > \epsilon) \leq \delta$$

Where:

- (h) = learned hypothesis
- (ϵ) = acceptable error
- (δ) = failure probability

So, A hypothesis is PAC-learned if:

- $\text{Error} \leq \epsilon$
- $\text{Probability of failure} \leq \delta$

Example

Suppose:

- $\epsilon = 0.05$
- $\delta = 0.01$

Interpretation:

- The learner can make at most 5% error.
- The probability that the learner fails to achieve this accuracy should be less than 1%.

That means,

- Error less than 5%
- Success probability 99%

The learner does not need perfect accuracy, only sufficiently good accuracy.

$$\text{Success Probability} = 1 - \delta$$

Advantages

- Provides theoretical foundation
- Measures learnability
- Helps compare algorithms
- Supports statistical learning

Limitations

- Strong mathematical assumptions
- Real-world data may not satisfy assumptions
- Sometimes computationally difficult

Introduction to Statistical Learning

Statistical learning is a **branch of Artificial Intelligence and Machine Learning** that **combines** concepts from **statistics, probability, and machine learning** to **analyze data and make predictions**.

- Statistical learning combines statistical techniques to **analyze and interpret data effectively**.
- It uses **probability theory** to handle **uncertainty and make predictions** based on data patterns.
- It also **incorporates machine learning methods** to enable systems to **learn automatically from data** without explicit programming.
- It focuses on: **Learning patterns from data** using statistical methods.

Goals

1. Prediction
2. Pattern recognition
3. Data analysis
4. Decision making

Statistical Learning Process

Statistical learning **follows a sequence of steps** through which **raw data is converted into useful predictions or decisions**.

The process involves **collecting data, training** a model, and **using the learned model** for prediction or analysis.

1. Data Collection

Data collection is the first step in the statistical learning process.

In this stage, **relevant data is gathered from different sources** such as databases, sensors, websites, surveys, or user interactions.

The **quality and quantity of data are very important** because the performance of the learning system depends heavily on the data provided.

Examples

- Collecting customer purchase records in an e-commerce system
- Gathering medical reports for disease prediction
- Recording weather information for forecasting

Importance

- Provides raw information for learning
- Helps identify patterns and relationships
- Better data leads to better model accuracy

2. Training Data

After data collection, the collected data is **prepared and organized into a training dataset**.

Training data consists of **examples that are used to teach the learning algorithm**.

In supervised learning:

- Training data contains both input and correct output values.

Example

Input	Output
Email contains “lottery”	Spam
Email from office	Not Spam

Functions of Training Data

- Helps the model learn patterns
- Used to estimate relationships between variables
- Enables prediction on unseen data

3. Learning Algorithm

A learning algorithm is **applied to the training data to discover patterns, rules, or relationships**.

The algorithm analyzes the data repeatedly and adjusts its internal parameters to **minimize prediction errors**.

Examples of Learning Algorithms

- Linear Regression
- Decision Trees
- Neural Networks
- Naive Bayes
- Support Vector Machines

Role of Learning Algorithm

- Learns from examples
- Identifies hidden patterns
- Builds mathematical relationships

Mathematical Representation

The algorithm tries to learn a function:

$$f : X \rightarrow Y$$

Where:

- X = input data
- Y = output prediction

4. Model Generation

After training, the algorithm **generates a model**.

A model is the **learned representation of patterns and relationships** present in the training data.

The generated model **can now make predictions for new unseen data**.

Example

A **trained spam filter model** learns:

Emails containing words like “offer” or “lottery” are likely to be spam.

Importance of Model

- Stores learned knowledge
- Used for prediction and decision-making
- Improves automatically with better training

5. Prediction / Decision

In the **final stage**, the trained model is used to make **predictions or decisions on new data**.

The model **receives unseen input data and predicts** the output based on learned patterns.

Examples

- **Predicting** house prices
- Detecting spam emails
- Recommending movies
- Diagnosing diseases

Example Flow

New Email → Trained Spam Model → Spam / Not Spam

Real-Life Example: Weather Prediction

Step	Activity
Data Collection	Collect temperature, humidity, wind data
Training Data	Historical weather records
Learning Algorithm	Weather prediction algorithm
Model Generation	Forecasting model
Prediction	Predict tomorrow's weather

Advantages of Statistical Learning Process

- Helps discover hidden patterns in data
- Enables automatic prediction and decision-making
- Improves system performance over time
- Useful in real-world AI applications

Applications

- Medical diagnosis
- Fraud detection
- Recommendation systems
- Stock market prediction
- Speech recognition
- Image classification

Introduction to Reinforcement Learning (RL)

Reinforcement Learning is a branch of Artificial Intelligence where an **agent learns how to behave in an environment by performing actions and receiving rewards or penalties.**

The goal of the agent is to learn a strategy (policy) that **maximizes the total reward over time.**

Working of Reinforcement Learning

1. Agent **observes** the current state.
2. Agent **selects** an action.
3. Environment **changes** state.
4. Agent **receives** reward/penalty.
5. Process **repeats** until goal is achieved.

Example

Robot Navigation

- A robot moves in a maze.
- Correct path → positive reward.
- Hitting wall → negative reward.
- After repeated trials, robot learns shortest path.

Characteristics of Reinforcement Learning

- Reinforcement learning is **based on trial-and-error learning**, where the agent learns by performing actions and observing their outcomes in the environment.
- In reinforcement learning, **rewards may sometimes be delayed**, meaning that the agent may receive feedback after a sequence of actions instead of immediately after every action.
- Reinforcement learning **does not require a human supervisor** or labeled training data because the agent learns automatically through interaction with the environment.
- It **involves sequential decision making**, where the current action chosen by the agent affects future states, actions, and rewards.
- The **agent gradually learns optimal behaviour** over time by continuously improving its actions in order **to maximize the total cumulative reward.**

Types of Reinforcement Learning

1. Passive Reinforcement Learning
2. Active Reinforcement Learning

Passive Reinforcement Learning

In **Passive Reinforcement Learning**, the agent follows a **fixed policy** and **learns how good the states are**.

The agent **does not decide actions**; **actions** are already **predefined**.

Objective

Evaluate the policy and estimate utilities of states.

Features of Passive RL

- Policy is fixed
- Agent only learns state values
- No exploration of new actions
- Simpler than active RL

Methods Used in Passive RL

1. Direct Utility Estimation

- In Direct Utility Estimation, the **utility values of states are estimated directly from the agent's experiences** in the environment.
- The **agent performs actions repeatedly** and **observes the rewards** received after reaching different states.
- The **average of the rewards obtained from multiple experiences** is calculated to estimate the utility of a particular state.
- **States that consistently provide higher rewards** are assigned **higher utility values**, while states with lower rewards receive lower utility values.
- **It -**
 - Directly estimates the utility (value) of states
 - Learns only from the actual rewards received
 - Does not build a model of the environment
 - If a state repeatedly leads to good outcomes, its utility becomes high.

Example

Suppose a robot reaches State A several times and receives rewards:

10, 12, 8, 10

Estimated utility:

$$U(A) = \frac{10 + 12 + 8 + 10}{4} = 10$$

The agent simply remembers that State A is usually worth about 10.

2. Adaptive Dynamic Programming (ADP)

- Adaptive Dynamic Programming is a reinforcement learning method in which the **agent learns the transition probabilities** and **reward model** of the environment through experience.
- The agent builds an internal model that describes how actions change the environment from one state to another.
- After learning the model, the agent applies dynamic programming techniques **to compute the optimal utilities** and policies for decision-making.
- ADP **combines learning and planning**, allowing the agent to improve its behavior based on both observed experiences and calculated future outcomes.
- the agent:
 - Learns a model of the environment
 - **Estimates:**
 - **Transition** probabilities
 - **Reward** function
 - Uses this model to compute utilities using dynamic programming methods
 - Instead of only remembering rewards, the agent learns:
 - **“If I take action a in state s, what next state will probably occur?”**
- The agent **creates an internal map of the environment** and plans using that model.

Example

The agent may learn:

$$P(S_2 | S_1, a) = 0.8$$

Meaning:

- From state S_1 ,
- after action a ,
- there is an 80% chance of reaching S_2 .

Then utilities are computed mathematically using Bellman equations.

A central update idea in ADP/value iteration is:

$$U(s) = R(s) + \gamma \sum_{s'} P(s' | s, a) U(s')$$

where:

- $R(s)$ = immediate reward
- γ = discount factor
- $P(s' | s, a)$ = transition probability

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3. Temporal Difference Learning (TD Learning)

- Temporal Difference Learning is a reinforcement learning technique in which the **agent learns directly from experience** without requiring a **complete model** of the environment.
- In TD learning, the **utility estimates are updated** after every step by **comparing** the **current prediction with the reward** and the **estimated utility** of the next state.
- The method **gradually improves utility values** by reducing the difference between successive predictions, known as the temporal difference error.
- TD learning combines the advantages of dynamic programming and direct learning from experience.

Characteristics of Temporal Difference Learning

- Temporal Difference Learning enables the agent to learn continuously from interactions with the environment.
- The utility estimates are updated immediately after each action, making learning faster and more adaptive.
- It does not require complete knowledge of transition probabilities or future rewards in advance.
- TD learning is widely used in robotics, game playing, and autonomous systems.

TD Update Formula

The utility of a state is updated using the following formula:

$$U(s) \leftarrow U(s) + \alpha[R + \gamma U(s') - U(s)]$$

Explanation of Terms

Where:

- $U(s)$ represents the current utility value of the present state s .
- α represents the learning rate, which controls how much the new experience influences the updated utility value.
- γ represents the discount factor, which determines the importance of future rewards compared to immediate rewards.
- R represents the immediate reward received after performing an action.
- $U(s')$ represents the estimated utility of the next state s' .

Working of TD Formula

- The agent first observes the reward obtained after moving to the next state.
- It then compares the current utility estimate with the improved estimate based on the reward and future utility.
- The **difference between these values** is called the temporal difference error.
- Finally, the utility value of the current state is adjusted to reduce this error and improve future predictions.

Example of Passive RL

Student Learning System

- Student follows fixed study schedule.
- System evaluates performance after each test.
- Learns which states produce better marks.

Active Reinforcement Learning

Definition

In **Active Reinforcement Learning**, the agent learns:

1. Which actions to take
2. How good the actions are

The agent must decide the best policy itself.

Features of Active RL

- No fixed policy
- Agent explores environment
- Learns optimal actions
- More complex than passive RL

Common Active RL Methods

1. Q-Learning

Q-Learning is a model-free reinforcement learning algorithm that **helps an agent learn the quality (Q-value) of actions** in different states.

It learns quality (Q-value) of actions.

The Q-value tells the agent:

“How good is it to take a particular action in a particular state?”

The agent learns this through:

- Trial and error
- Rewards and penalties
- Continuous interaction with the environment

The goal is to learn the **optimal policy** that **maximizes long-term rewards**.

Q-Learning Formula

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Meaning of Each Term

Symbol	Meaning
$Q(s, a)$	Current Q-value
α	Learning rate (how much new learning matters)
R	Immediate reward received
γ	Discount factor for future rewards
s'	Next state
$\max Q(s', a')$	Best possible future Q-value

Advantages

- Simple
- Widely used
- Model-free

3. SARSA

SARSA stands for: (State, Action, Reward, Next State, Next Action)

Updates values using the action actually taken.

SARSA is a reinforcement learning algorithm that **learns the value of actions** based on the **action actually taken by the agent**.

It is an **on-policy learning algorithm**, meaning:

- The agent learns from the same policy it follows while interacting with the environment.

SARSA helps the agent learn:

- Which actions are **good**
- Which actions are **risky**
- How to maximize long-term rewards

SARSA Formula

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma Q(s', a') - Q(s, a)]$$

Meaning of Each Term

Symbol	Meaning
$Q(s, a)$	Current Q-value
α	Learning rate
R	Immediate reward
γ	Discount factor
$Q(s', a')$	Q-value of the next state and actual action taken

Applications of Reinforcement Learning

Application	Description
Robotics	Robot movement and control
Gaming	Chess, Go, video games
Self-driving Cars	Decision making while driving
Recommendation Systems	Personalized suggestions
Healthcare	Treatment planning
Finance	Stock trading strategies

Advantages of Reinforcement Learning

- Learns automatically
- Adapts to environment

- Handles complex problems
- Useful for sequential tasks

Limitations of Reinforcement Learning

- Requires large training time
- High computational cost
- Exploration may be risky
- Difficult reward design